Requirements: Project Orion Mailing Solutions

Jonathan Wong
Abrar Zahin
Joshua Ooroth
Frankie Cacio
Aaryan Kumar

Department of Computer Science and Engineering
University of South Florida
Tampa, FL 33620

Email:
jcwong@usf.edu
abrarz@usf.edu
jooroth@usf.edu
fcacio@usf.edu
aaryankumar@usf.edu

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May 1th, 2026

History of document

- **Version 1.0.0** (February 6, 2026) – Initial document created
- **Version 1.0.1** (February 19, 2026) – Simplified and updated requirements, specified glossary and assumptions
- **Version 1.0.2 (**May 1, 2026) – Final revision

Table of Contents

1. Introduction

The problem addressed by these requirements is: "Can we design and build an automatic continuous assembly line synthetic handwritten letter generator and envelope packing for less than $500 preferably less than $300 that cannot be detected as fake by at least 90% of people."

2. Glossary

1) *Automatic*: a production system that will produce a fully packaged envelope with at most one human intervention per production cycle

2) *Handwriting*: letters written in the English language that is written with pen or pencil

3) *Continuous assembly line*: A workflow where input items move through different processing stations in a repeatable sequence, operating continuously, producing complete ready-to-mail envelopes at the output

4) *Synthetic Handwriting*: Text generated by a machine using pen-based strokes intended to resemble human-written English characters

5) *Letter*: A 3x5 inch index card

6) *Envelope*: a packaged and sealed package with one 3x5inch index card with a stamp and address

3. Assumptions

1) Pen plotter utilizes the GRBL firmware for plotting.

2) Text written on the letter will be generated through human input

3) Input materials (i.e. index cards, envelopes, and stamps) will be loaded by human

4) 3in x 5in Index cards and Standard sized 10.4cm x 24.1cm Envelopes are used

5) One human overseer over the process

4. Requirements

1) As a user, I want the full handwriting and packaging process to require no more than one human interaction after the index card and envelope are loaded, so that I can complete other tasks while the envelope is being processed (Partially Complete)

2) As a user, I want the handwriting system to automatically write the selected message on the index card and envelope, so that I do not have to manually perform the writing step (Complete)

3) As a user I want the index card and the envelope to look human written with a real pen so that 85% of people will not be able to detect that it was written by a robot (Complete with 86.3%)
   Acceptance Criteria:
   a. Handwriting includes variation in stroke, spacing, or alignment
   b. At least 85% of test observers do not identify the letter as machine-generated

4) As a user, I want each envelope and index card to be written in a unique handwriting style so that no two handwritten outputs look identical across at least 10,000 jobs (Complete)

5) As a user, I want the packaging system to automatically insert the completed index card into the envelope, so that the final envelope is ready for pickup without requiring manual packaging (Partially Complete)

6) As a project sponsor, I want to be able to produce this assembly line with less than $500, preferably under $300 so that I can justify designing this assembly line by college students (Complete with $481.55)

7) As a user I want the assembly line system to be modular, so that I can separate, transport, and reassemble the system without additional assistance (Complete)

5. Needs and factors

5.1 Public health needs

Public health needs are not applicable to this project. The system operates in a controlled environment and does not directly affect human health or medical outcomes.

5.2 Public safety needs

Public safety needs are not applicable to this project. While the system contains moving mechanical components, it is intended for supervised indoor use and does not pose significant risk to the public when operated as designed.

5.3 Public welfare needs

Public welfare needs are not applicable to this project.

5.4 Global factors

Global factors are not applicable to this project. The system is designed for small-scale, localized use and is not intended for global deployment or international infrastructure

5.5 Cultural factors

See 5.6 Social factors.

5.6 Social factors

Social factors are not applicable to this project. Cultural and social needs are addressed by the requirement (2) for the user. Requirement (2) notes that the system writes mimicking natural human handwriting, satisfying the social and cultural expectations for handwritten greeting cards.

5.7 Environmental factors

Environmental factors are not applicable to this project. The system consumes paper, ink, and electrical power, but its environmental impact is comparable to standard manual letter writing and packaging.

5.8 Economic factors

Economic factors are addressed by requirement (3) for the user. Requirement (3) notes the need for the assembly line cost to be less than $500, preferably under $300.

References

[1] "What Is the Most Popular Type of Greeting Card?," Pynk Cards, Sep. 4, 2024 [Online]. Available: https://pynk.net/blogs/news/what-is-the-most-popular-type-of-greeting-card. [Accessed: Feb. 6, 2026].

Specifications: Orion Mailing Solutions

Jonathan Wong

Abrar Zahin

Joshua Ooroth

Frankie Cacio

Aaryan Kumar

Bellini College of AI, Cybersecurity and Computing

University of South Florida

Tampa, FL 33620

Email:

jcwong@usf.edu

abrarz@usf.edu

jooroth@usf.edu

fcacio@usf.edu

aaryankumar@usf.edu

Version v1.0.0

3/6/2026

History of document

- **Version 1.00** (3/6/2026)

Table of Contents

1. Introduction

AI is slowly phasing out human labor returning free time from repetitive tasks, however there is a loss in some human aspects of life, such as handwriting, that carry personal meaning and emotional connection. To solve this, "Can we design and build an automatic continuous assembly line synthetic handwritten letter generator and envelope packing for less than $500 preferably less than $300 that cannot be detected as fake by at least 90% of people."

2. Glossary

1) *Recurrent Neural Networks (RNNs)*: are a class of artificial neural networks designed for sequential or time-series data, featuring feedback loops that allow previous information to persist

2) *Geometric code (G-code)*: is the primary programming language used to control automated machine tools such as CNC machines and 3D printers

3) *Pen Plotter*: Writing utensil device which uses g-code to write

4) *Automatic*: a production system that will produce a fully packaged envelope with at most one human intervention per production cycle

5) *Handwriting*: letters written in the English language that is written with pen or pencil

6) *Continuous assembly line*: A workflow where input items move through different processing stations in a repeatable sequence, operating continuously, producing complete ready-to-mail envelopes at the output

7) *Synthetic Handwriting*: Text generated by a machine using pen-based strokes intended to resemble human-written English characters

8) *Letter*: A 3x5 inch index card

9) *Envelope*: a packaged and sealed package with one 3x5inch index card with a stamp and address

3. Constraints

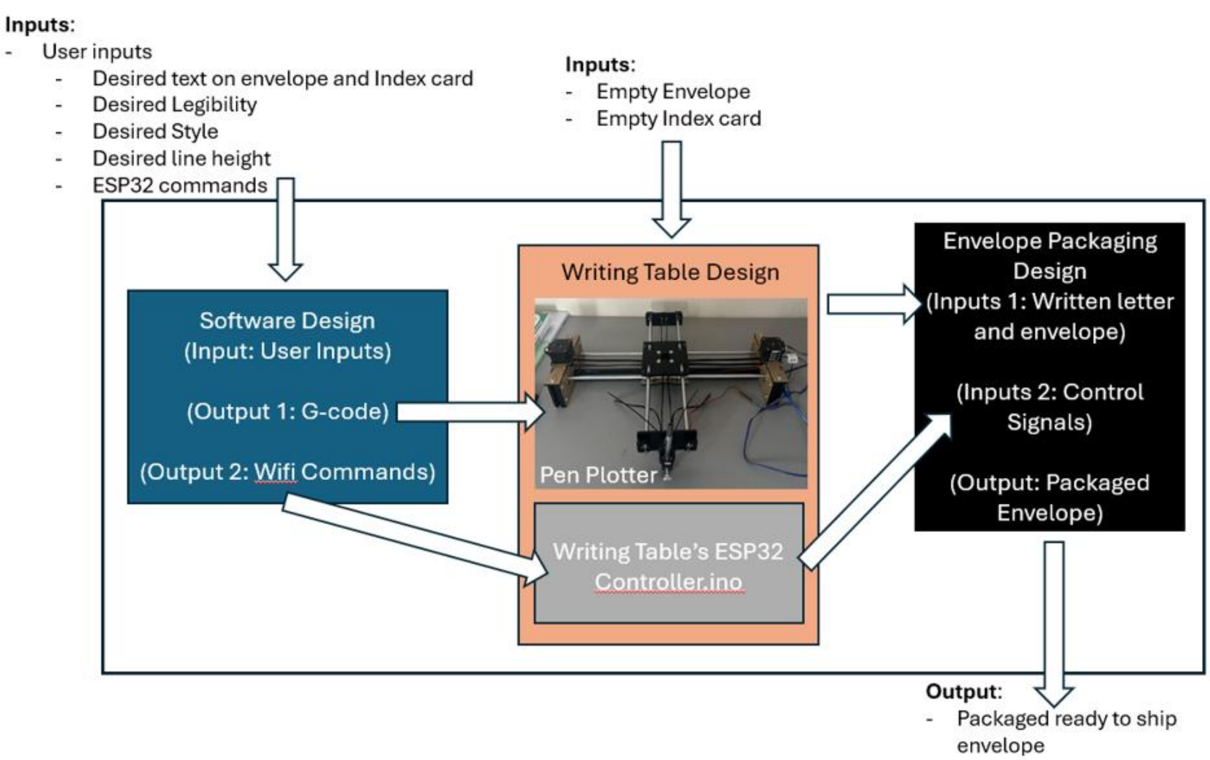
- **Cost:** Total system cost <= $500
    - Target Cost: <= $300
- **Human Interaction:** Maximum one human action per cycle
- **Uniqueness:** No two handwriting outputs may be identical
- **Physical Medium:** Must use real pen on standard #10 envelope and 3x5 index card
- **Functionality:** Must function as a continuous assembly line process

4. Applicable standards

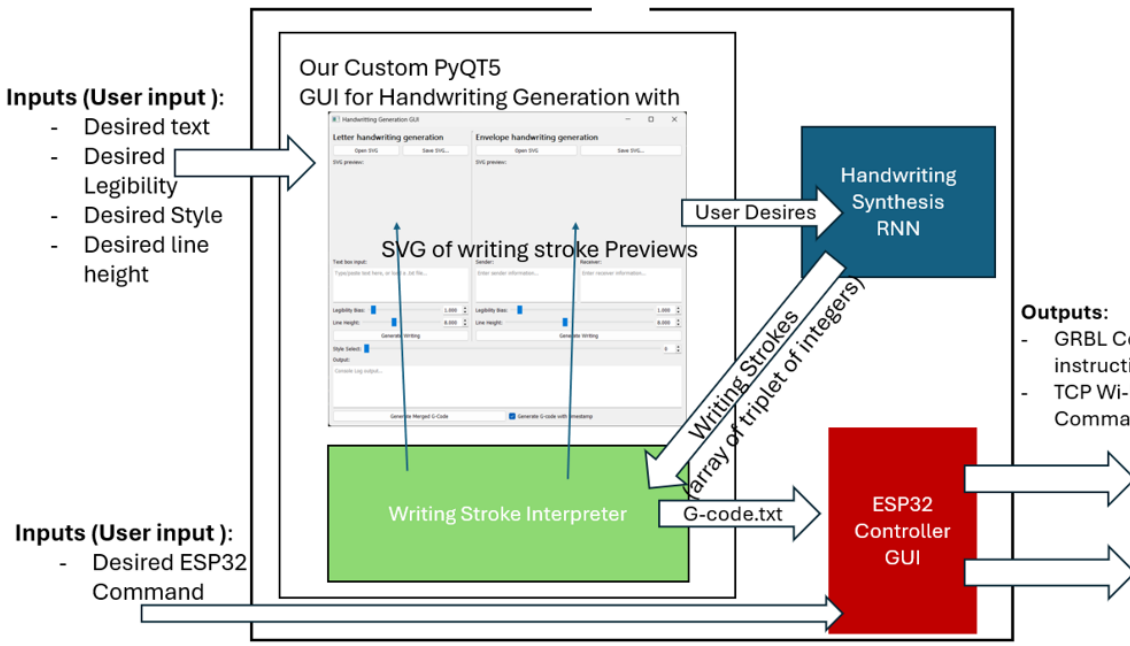
- **G-code:** Standard language for CNC machine code
- **USPS Mailing Standards:** Envelope and postage compliance

5. Design

5.1 Overall design:



5.2 Software design:



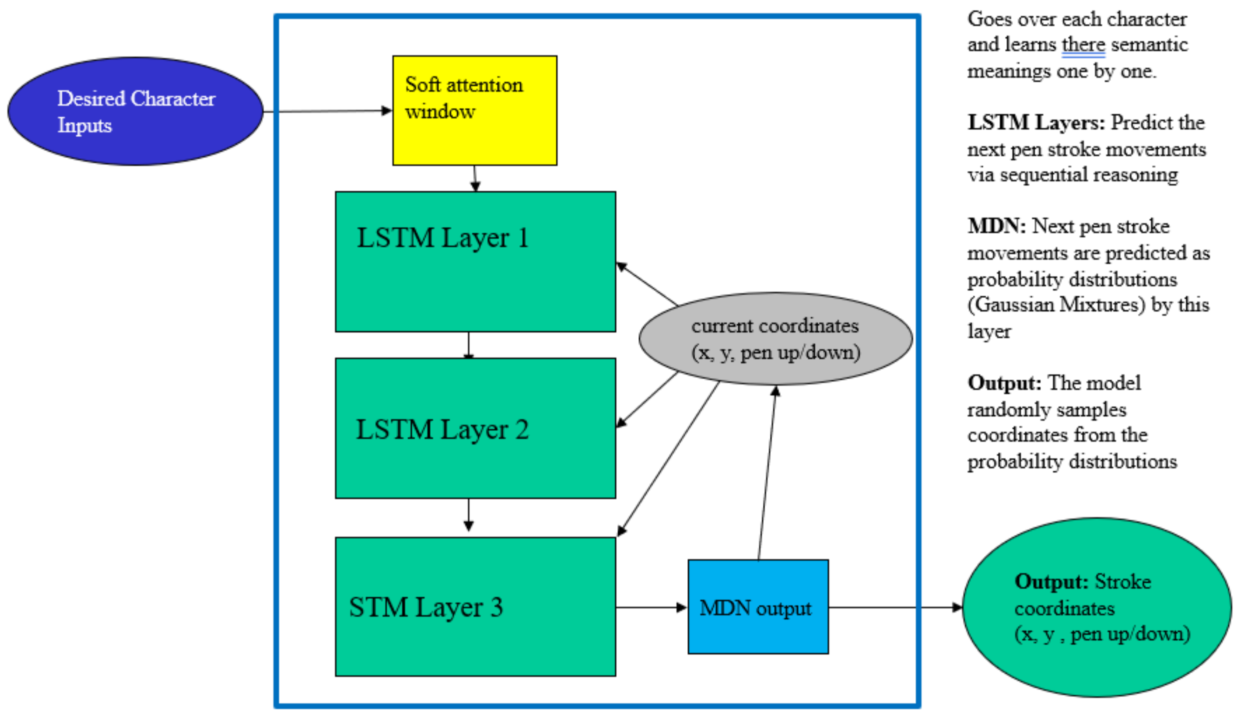
5.2.1 Software design – Handwriting Synthesis RNN:

Soft Attention Window: Goes over each character and learns there semantic meanings one by one.

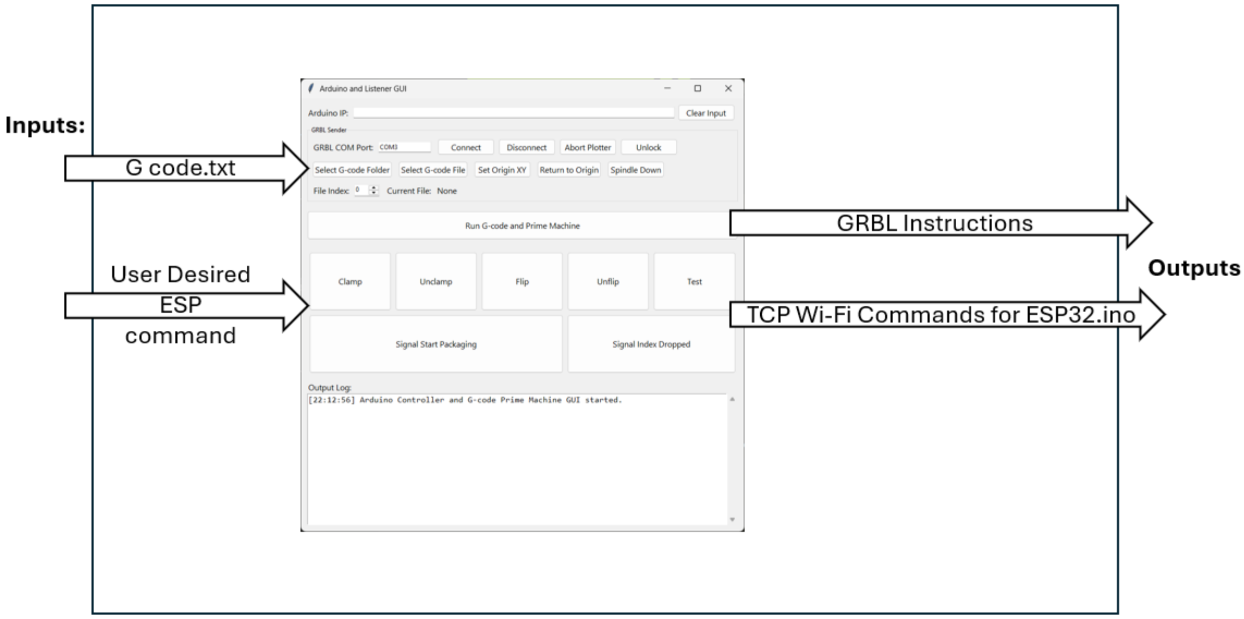
LSTM Layers: Predict the next pen stroke movements via sequential reasoning

MDN: Next pen stroke movements are predicted as probability distributions (Gaussian Mixtures) by this layer

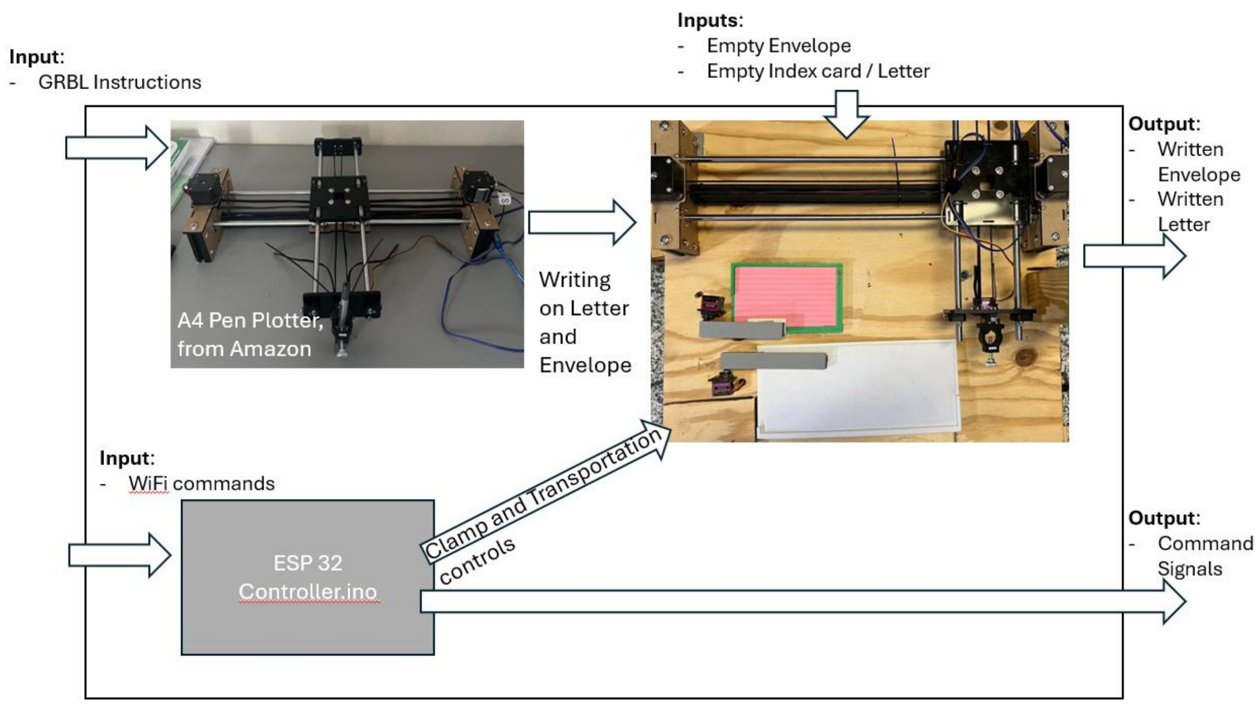
Output: The model randomly samples coordinates from the probability distributions



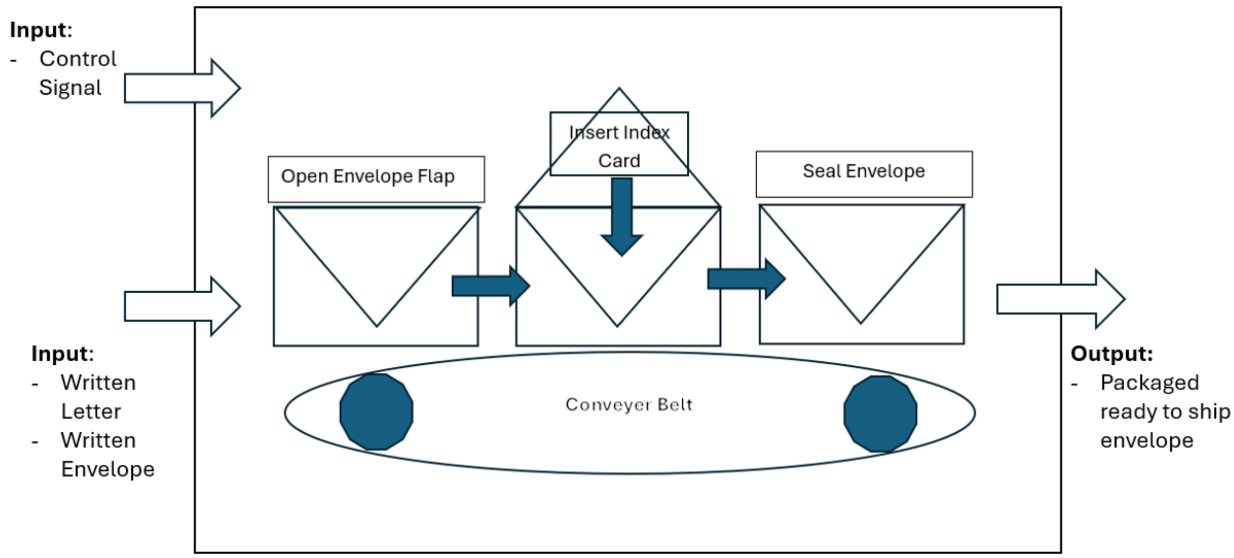
5.2.2: Software design - ESP32 GUI:



5.3 Pen Plotter Platform design



5.4 Envelope Packaging System design:



6. Risk analysis and mitigation

- **Shipping Delays and Ordering miscommunication**: When shipping mechanical parts (such as motors) are ordered incorrectly and/or having shipping delays
    - Mitigation: 3D printing, printing our own pieces or temporary solutions to prevent and/or continue development while shipping is slowed
    - Mitigation: Amazon Prime, so delivery will be faster than standard
    - Mitigation: Draft up documents for parts and links directly to product on Amazon for Cristian (Our project sponsor) to buy from
- **Handwriting Detection Risk:** Output may not pass 90% human test
    - Mitigation: Iterative user testing & parameter tuning
- **Mechanical Jamming:** Envelope or index card getting caught in transit between stations
    - Mitigation: Guided rails and alignment fixtures
    - Mitigation: Rapid prototyping leveraging 3d printer to test new parts as we design them
- **User Misinterpretation:** Misplacement of input materials, (i.e. envelope and index cards)
    - Mitigation: including guide markers in design to prevent misalignment of materials

7. Design trade-offs

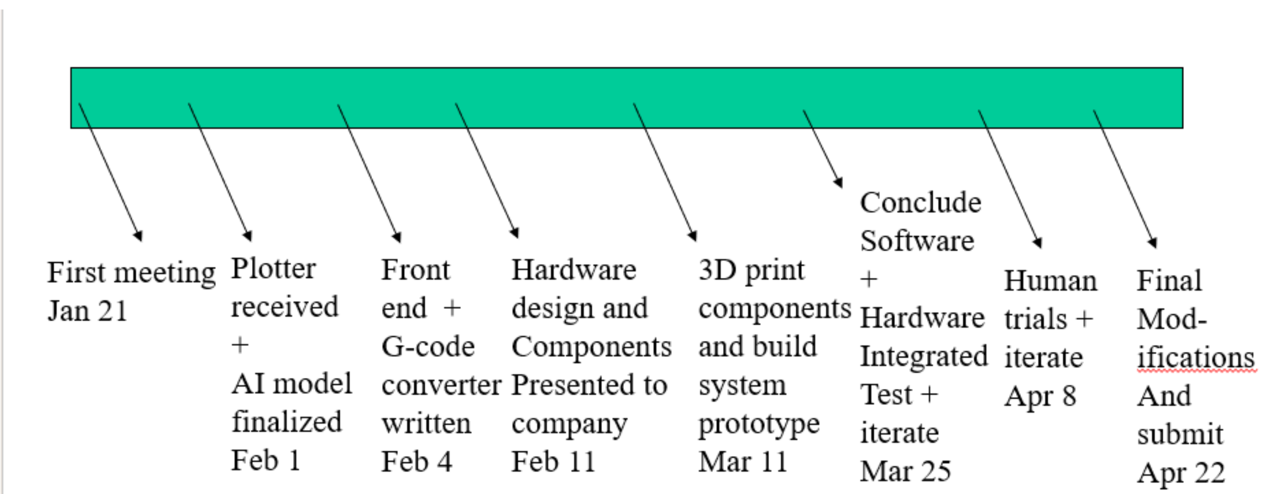
Transfer mechanism trade-off:

The original design for transferring envelopes from the pen plotter to the assembly line used a suction cup system to pick up and move the envelope. While this approach would allow precise placement, it required additional components such as a vacuum pump and control hardware, increasing both cost and implementation time. To reduce system complexity and stay within project constraints, the final design instead uses a motor-driven lever mechanism to lift the platform and allow the envelope to slide into the assembly line using gravity.

Stamping application trade-off:

Initially, the design placed all envelope processing steps after the pen plotter in a single assembly line. However, this approach created an issue because the envelope would be flipped during the assembly process, making it difficult to apply the stamp on the correct writing surface. To resolve this, the stamping step was moved to the beginning stage immediately after the pen plotter completes writing. This change ensured proper stamp placement and influenced the design of the transfer system, which now flips the envelope as it moves into the assembly line.

8. Project plan



9. Traceability to requirements

Requirement	**Description**	**Does our design fully cover the requirements?**	**Does our design partially cover the requirements?**	**Does our design exceed this requirement?**
Undetectable handwriting generation	As a reader of the letter I want to read a letter that looks human so that I do not realize that it was written and packaged by a robot	yes		
Continuous writing on multiple cards	As a greeting card maker, I want the automated system to write on many cards at a fast pace so that I can save time and labor cost in production.	yes		
Automate envelope packaging	As a user I want an automatic envelope stuffer so that generated letters can be packed autonomously with minimal human interaction.	yes		
Automate stamping envelopes	As a distributor I want the automated system to stamp envelopes so that the envelopes are ready for shipment when I get them	yes		
Pricing	As a patron I want to be able to produce this		Yes (the total expenses is under $500)	

	assembly line with less than $500, preferably under $300 so that I can justify making this assembly line using college students and not real engineer			

10. Traceability to needs and factors

Needs and factors	**Does our design fully cover the needs?**	**Does our design partially cover the needs?**	**Is the need applicable to our project?**	**Description**
Public health			No	
Public safety	Yes			While the system contains moving mechanical components, it is intended for supervised indoor use and does not pose significant risk to the public when operated as designed.
Public welfare needs			No	
Global factors			No	The system is designed for small-scale, localized use and is not intended for global deployment or international infrastructure
Environmental factors	Yes			The system consumes paper, ink, and electrical power, but its environmental impact is comparable to

				standard manual letter writing and packaging.
Social factors and cultural factors	Yes			The goal of the design is to meet social and cultural requirements.
Economic factors	Yes			The goal of the design is to spend under $500

References

1. J. Wong, A. Zahin, J. Ooroth, F. Cacio, and A. Kumar, "Requirements: Project Orion Mailing Solutions," Department of Computer Science and Engineering, University of South Florida, Tampa, FL, USA, Version 1.00, Feb. 6, 2026.
2. S. Gordon, "GRBL v1.1 G-Code Configuration and Command Guide," Open-source documentation, 2019. [Online]. Available: https://github.com/gnea/grbl
3. United States Postal Service, "Domestic Mail Manual (DMM)," USPS, Washington, DC, USA, 2026. [Online]. Available: https://pe.usps.com

Test Plan: Orion Mailing Solutions

Jonathan Wong

Abrar Zahin

Joshua Ooroth

Frankie Cacio

Aaryan Kumar

Bellini College of AI, Cybersecurity and Computing

University of South Florida

Tampa, FL 33620

Email:

jcwong@usf.edu

abrarz@usf.edu

jooroth@usf.edu

fcacio@usf.edu

aaryankumar@usf.edu

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History of document

- **Version 1.00** (date) – Initial document created
- **Version 1.01** (3/24/2026) – Correction in Success Criteria section of Design of test case, all 45 cases needs to be successful in a row in order to with 90% confidence the success rate is at least 95%

Table of Contents

1. Introduction

AI is slowly phasing out human labor returning free time by automating repetitive tasks, however there is a loss in some human aspects of life, such as handwriting, that carry personal meaning and emotional connection which printed text lack. To solve this, “Can we design and build an automatic continuous assembly line synthetic handwritten letter generator and envelope packing for less than $500 preferably less than $300 that cannot be detected as fake by at least 90% of people”.

2. Glossary

10) *Recurrent Neural Networks (RNNs)*: are a class of artificial neural networks designed for sequential or time-series data, featuring feedback loops that allow previous information to persist

11) *Geometric code (G-code)*: is the primary programming language used to control automated machine tools such as CNC machines and 3D printers

12) *Pen Plotter*: Writing utensil device which uses g-code to write

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15) *Continuous assembly line*: A workflow where input items move through different processing stations in a repeatable sequence, operating continuously, producing complete ready-to-mail envelopes at the output

16) *Synthetic Handwriting*: Text generated by a machine using pen-based strokes intended to resemble human-written English characters

17) *Letter*: A 3x5 inch index card

18) *Envelope*: a packaged and sealed package with one 3x5inch index card with a stamp and address

3. Parameters to be tested

List what you will test here.

1. Unique Handwriting Generation
    i. Must generate and print handwriting that is repeatably legible and completely unique.
2. Handwriting traceability
    i. pass as human against 90% of test subjects
    ii. initial number of test subjects will be 50 people (as requested by Cristian Dutescu, Orion Defense Solutions)
3. Writing repeatability, successfulness in continuous writing of multiple cards and envelopes
4. Packaging repeatability, successfulness in continuous packaging of multiple cards and envelopes
    i. Envelope gets stamped placed on the front
    ii. Envelope gets flipped and placed into packaging system from writing station
    iii. Envelope able to moves to next stations
    iv. Opening Envelope flap
    v. Packaging the envelope
    vi. Packaged envelope is outputted
5. Envelope and letter placement
    i. Envelopes and letters must be placed consistently in the same place, so the text is plotted in the correct positions every time.
6. System timing
    i. Each stage of the system should pass off to one another continuously, where the next stage only begins once the previous stage is fully complete

4. Design of testing

Describe the design of your testing here. A table could be one way to do this

Desired success rate is 95%. Using Rule of Three to find number of trails to say with 90% confidence, the true success rate is at least 95%

$$n \geq \frac{\ln(1 - C)}{\ln(R)}$$

n = # of trails

C = desired confidence level

R = target success rate

$$n \geq \frac{\ln(1 - 0.90)}{\ln(0.95)}$$

Solving for n, n equals approx. 44.9, rounding up to 45
so with 45 trials and no errors, then with 90% confidence the success rate is at least 95%.

Initial number of test subjects will be 50 people (Sample size approved by Cristian Dutescu, Orion Defense Solutions)

Parameter	**Testing Method**	**Measurement Method**	**Success Criteria**
Unique handwriting generation	Repeatably plot the same sentence	Visually compare each test sample	Each test 45 sample are unique to ensure with 90% confidence the success rate of uniqueness is at least 95%
Handwriting Traceability	Plot a sentence and collect 50 human samples of the same sentence	Conduct a survey where people compare the samples	AI generated sample to pass as a human sample to 90% of people surveyed
Writing repeatability	Generate 45 writing examples	Visually review examples for errors in: - No writing - Missing characters - Excessive overlapping writing	All 45 generated write successfully with no major errors that effect readability to ensure with 90% confidence the success rate is at least 95%
Packaging repeatability	45 trails of placing an empty envelopes and letters into the packaging system	A successful trail must contain:	All 45 packaged letter in the stamped envelope to ensure with 90% confidence

	producing packaged envelopes	i. Envelope gets stamped placed on the front ii. Envelope gets flipped and placed into packaging system from writing station iii. Envelope able to moves to next stations iv. Opening Envelope flap v. Packaging the envelope vi. Packaged envelope is outputted	the success rate is at least 95%
Envelope and letter placement	Place 45 envelope and letter into designated areas and plot after clamping them down	Running the pen plotter marks envelope and letter start position in a consistent spot	All 45 Letter plus sender and receiver information is plotted in the correct positions to ensure with 90% confidence the success rate is at least 95%

5. Test case #1

A) Handwriting Authenticity Test
B) System Configuration
   i. Pen plotter connected
   ii. Index card loaded
   iii. Predefined sentence used for testing
C) Input
   i. Text input: "Dear friend, I hope this letter finds you well."
   ii. Repeat text input each with its own unique handwriting style 10 times
   iii. Present the 10 handwriting styles to 50 human test subjects
   iv. Ask each subject: "Do you believe this handwriting was written by a human?
   v. Collect and review Yes or No survey data
D) Expected Output
   i. 10 distinct handwriting styles
   ii. At least 90% of test subject identify writing as human written for each style
   iii. Each sample is legible with no missing characters

6. Test case #2

A) Full System Integration Test
B) System Configuration
   i. Pen plotter connected
   ii. Clamp installed for both envelope and index card
   iii. Transfer mechanism connected successfully between writing platform and assembly line
   iv. Envelope stuffing assembly line completed and operational
   v. Full system timing sequence programmed
C) Input
   i. Load one standard #10 envelope into designated position
   ii. Load one 3x5 index card into designated position
   iii. Input predefined text into system
   iv. Full system process begins and should complete the following sequences
      - Clamp card and envelope
      - Plot text
      - Apply stamp
      - Release clamp
      - Transfer envelope
      - Open envelope and insert index card
      - Seal envelope
   v. Repeat full process for 45 trials
D) Expected output
   i. System completes full sequence without interruption
   ii. Text and stamp positioning is accurate
   iii. Index card is stuffed and sealed in envelope
   iv. 45 trials complete without mechanical or sequencing errors

References

OpenAI. ChatGPT conversation on sample size and confidence intervals.
URL: https://chatgpt.com/share/69b46cb9-8e54-8011-aad4-8985bc8b2290.
Last accessed: March 13, 2026.

Design Review Slides

Orion Mailing Solution

Jonathan Wong
Abrar Zahin
Joshua Ooroth
Frankie Cacio
Aaryan Kumar

Bellini College of AI, Cybersecurity and Computing
University of South Florida
Tampa, FL 33620

Acknowledgments

- **Cristian Dutescu (Orion Defense Solutions)**
- **Dr. Ken Christensen (University of South Florida)**
- **Alex Grave's Github repository for the AI model:** https://github.com/sjvasquez/handwriting-synthesis
- **Youtube video showcasing the inspiration of mechanics of an envelope stuffing machine:** https://youtu.be/o60orGjWthE?si=vfvJxoXH5-omNMsm

Agenda

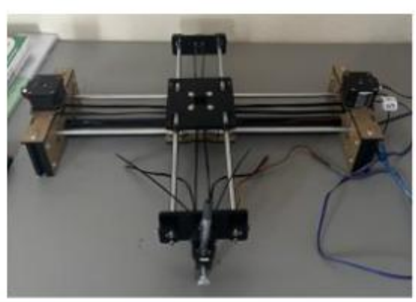
- **Background**
- **Problem**
- **Requirements**
- **Design**
- **Constraints**
- **Applicable standards**
- **Risk analysis and mitigation**
- **Project plan**

Background

- **Recurrent Neural Networks (RNNs):** are a class of artificial neural networks designed for sequential or time-series data, featuring feedback loops that allow previous information to persist

- **Geometric code (G-code):** is the primary programming language used to control automated machine tools such as CNC machines and 3D printers

- **Pen Plotter:** Writing device which uses G-code to write



Problem

- **Can we design and build an automatic continuous assembly line that can write on each envelope with a unique handwriting style, insert an index card, and stamp each envelope for mailing with minimal human involvement (one human in the loop) for less than $500, preferably less than $300 that cannot be detected as machine generated by at least 90% of people?**

Requirements

1. As a user I want the system to be automated with at most 1 human interaction between the start (putting in the index card and envelope) and the end (picking up the packaged envelope) so that I will be able to save time and be able to do other tasks as the envelope is being processed

2. As a user I want the index card and the envelope to look human written with a real pen so that 90% of people will not be able to detect that it was written and packaged by a robot

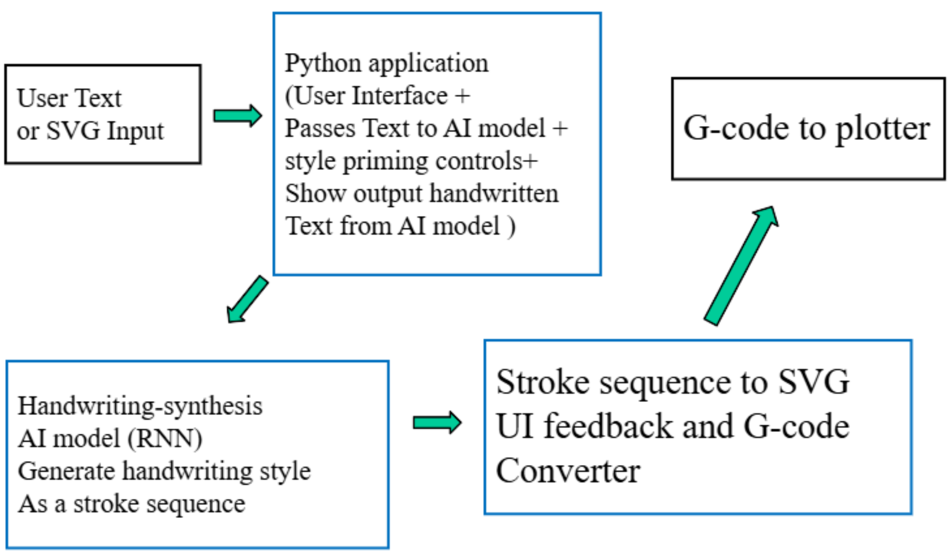
3. As a user, I want each envelope to be written with a unique handwriting so that each envelope is distinguishable from one another.

Requirements cont.

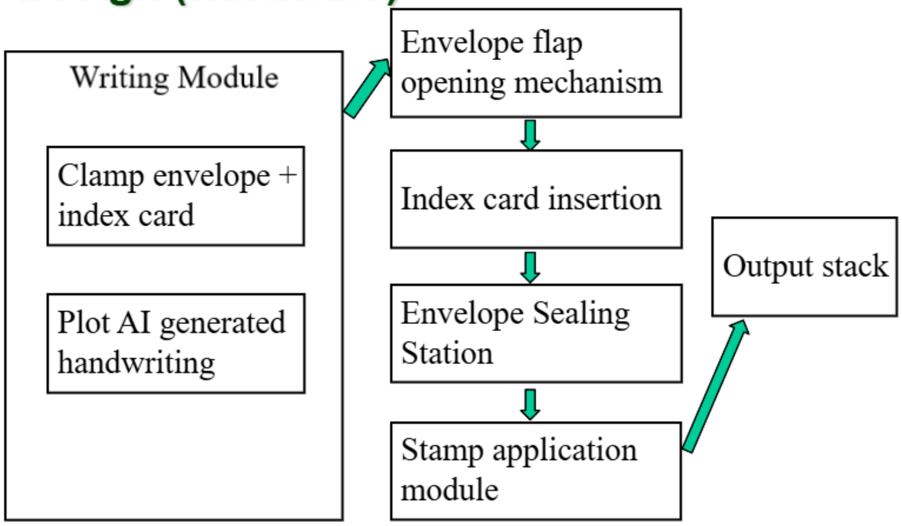
4. As a project sponsor I want to be able to produce this assembly line with less than $500, preferably under $300 so that I can justify designing this assembly line using college students and not real engineers

5. As a user I want this assembly line system to be as self-contained as possible so that the machine is portable and recreated

Design (Software)



Design (Hardware)



Design (parts)

- Writing System: Pen plotter with motor controllers
- Software: Neural Network model which includes text input interface, handwriting generator, and g-code converter
- Envelope packing sub-system: 3d printed parts, Vacuum system, stamp placement mechanism, conveyer belt system

Constraints

- **Cost:** Total system cost ≤ $500
  - Target Cost: ≤ $300
- **Human Interaction:** Maximum one human action per cycle
- **Uniqueness:** No two handwriting outputs may be identical
- **Physical Medium:** Must use real pen on standard #10 envelope and 3x5 index card
- **Functionality:** Must function as a continuous assembly line process

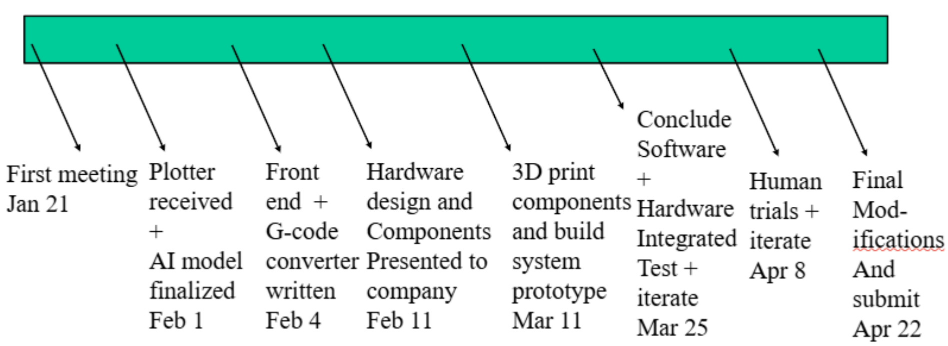
Applicable standards

- **G-code:** Standard language for pen plotter
- **USPS Mailing Standards:** Envelope and postage compliance

Risk analysis and mitigation

- **Handwriting Detection Risk:** Output may not pass 90% human test
  - Mitigation: Iterative user testing & parameter tuning
- **Mechanical Jamming:** Envelope or index card getting caught in transit between stations.
  - Mitigation: Guided rails and alignment fixtures
  - Mitigation: Rapid Prototyping leveraging 3d printer to test new parts as we design them
- **User Misinterpretation:** Misplacement of input materials, (i.e. envelope and index cards)
  - Mitigation: including guide markers in design to prevent mis alignment of materials

Project plan



Final Design Slides

Orion Mailing Solution

•Jonathan Wong
•Abrar Zahin
•Joshua Ooroth
•Frankie Cacio
•Aaryan Kumar

Bellini College of AI, Cybersecurity and Computing
University of South Florida
Tampa, FL 33620
email@cse.usf.edu

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- **Dr. Ken Christensen (University of South Florida)**
- **Alex Grave's Github repository for the AI model:** https://github.com/sjvasquez/handwriting-synthesis
- **AX4 Pen Plotter Writing Robot Open Source DIY Kit Corexy A4 Size Learn Programming:** Bought from Amazon.com from **IKRANBIRD** store

Acknowledgments

- **Youtube video showcasing the inspiration of mechanics of an envelope stuffing machine:** https://youtu.be/o60orGjWthE?si=vfvJxoXH5-omNMsm

Agenda

- **Background**
- **Problem**
- **Requirements**
- **Demo**
- **Customer statement**
- **Design**
- **Constraints**
- **Applicable standards**
- **Trade-offs made**

Background

- **AI is phasing out human labor in many tasks**
  - Cost is saved
  - Redundant tasks can be avoided, returning time to humans
- **Loss in some human aspects**
  - Handwriting carries personal meaning and identity
  - There is an emotional factor to human handwriting
  - Printed writing is perceived as lifeless
- **Project Vision**
  - To fully automate handwritten communication while preserving the personal, emotional, and human-like qualities that make handwriting meaningful

Problem

Can we design and build an automatic continuous assembly line that can write on each envelope with a unique handwriting style, insert an index card, and stamp each envelope for mailing with minimal human involvement (one human in the loop) for less than $500, preferably less than $300 that cannot be detected as machine generated handwriting by at least 85% of people?

Requirements

1. As a user, I want the full handwriting and packaging process to require no more than one human interaction after the index card and envelope are loaded, so that I can complete other tasks while the envelope is being processed (Partially Complete)

2. As a user, I want the handwriting system to automatically write the selected message on the index card and envelope, so that I do not have to manually perform the writing step (Complete)

3. As a user I want the index card and the envelope to look human written with a real pen so that 85% of people will not be able to detect that it was written by a robot (Complete with 86.3%)

4. As a user, I want each envelope and index card to be written in a unique handwriting style so that no two handwritten outputs look identical across at least 10,000 jobs (Complete)

Requirements (cont.)

5. As a user, I want the packaging system to automatically insert the completed index card into the envelope, so that the final envelope is ready for pickup without requiring manual packaging (Partially Complete)

6. As a project sponsor, I want to be able to produce this assembly line with less than \$500, preferably under \$300 so that I can justify designing this assembly line by college students (Complete with \$481.55)

7. As a user I want the assembly line system to be modular, so that I can separate, transport, and reassemble the system without additional assistance (Complete)

Demo

- **Handwriting synthesis GUI**
- **Handwriting Demo using synthesis GUI**
- **Survey Results of AI detectability**
- **Mechanical packaging demo**
- **Individual automated stations**

Demo (Completes req. 4)

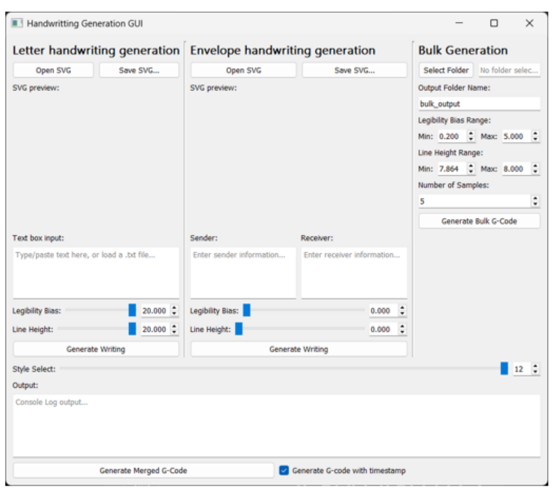
Legibility Slider
- Min:0.000
- Max: 20.000

Line Height Slider
- Min:0.000
- Max: 20.000

13 different styles

Each Slider has 20,001 possibilities

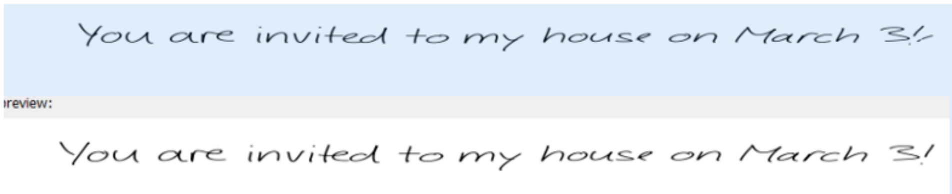
Totaling: $20{,}001^2 * 13$
= 5,200,520,013



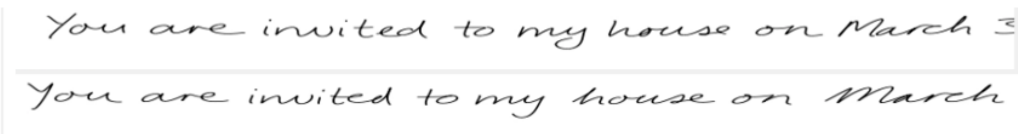
OVER 5 BILLION possibilities

Demo (Completes req. 4)

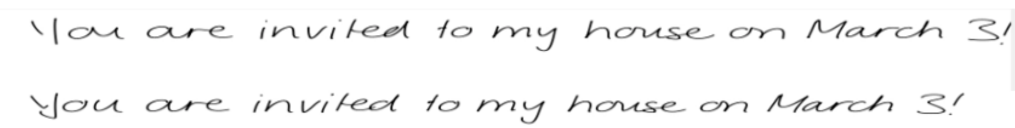
Style 3 (same styles, small legibility difference , different outputs):



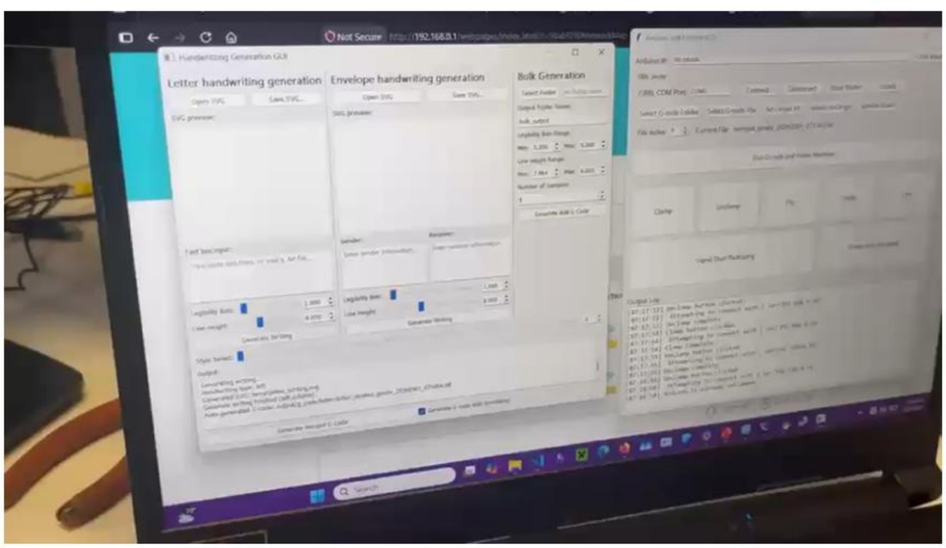
Style 5:



Style 8:

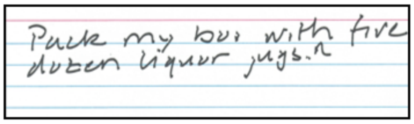


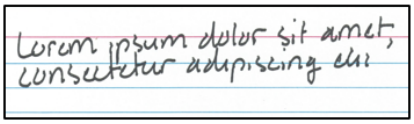
Demo (Completes req. 2 and show 4)

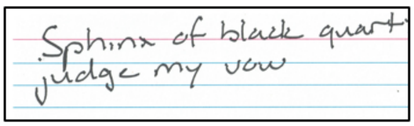


Demo (Completes req. 3)

- **Survey 1 (77 responses)**
  - Raw AI Pass Rate: 59.3%
  - Human False Positive: 23.4%
  - Youden's J: 17.4%
  - False Positive Adjusted Pass Rate: 82.6%
- **Survey 2 (27 Responses)**
  - Raw AI Pass Rate: 64.7%
  - Human False Positive: 29.4%
  - Youden's J: 5.5%
  - False Positive Adjusted Pass Rate: 94.5%





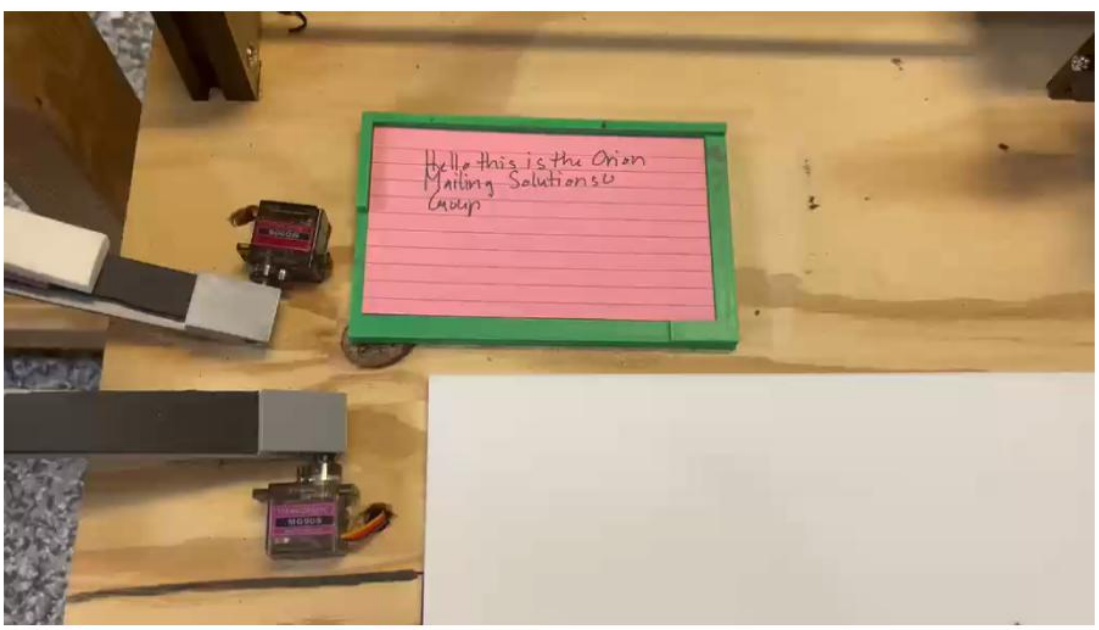


- **Combined Results**
  - False Positive Adjusted Pass Rate: 86.3%

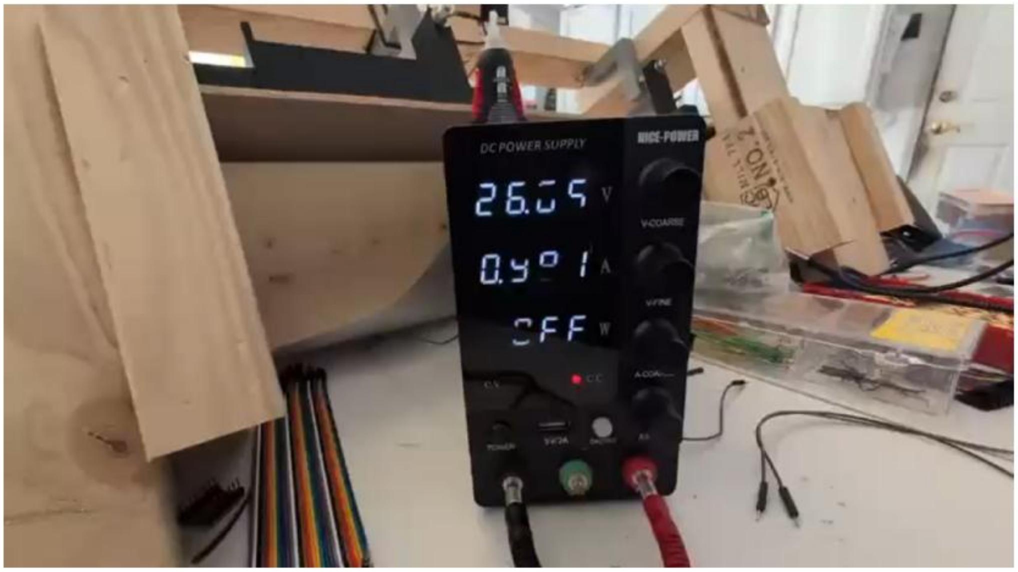
Youden's J Statistic:

True positive rate – false positive rate

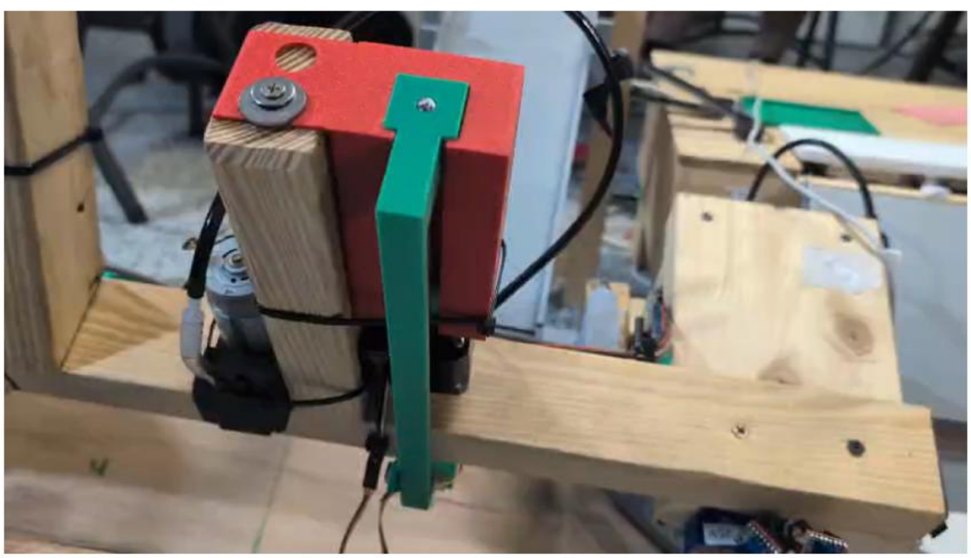
Demo (Partial Complete req. 5)



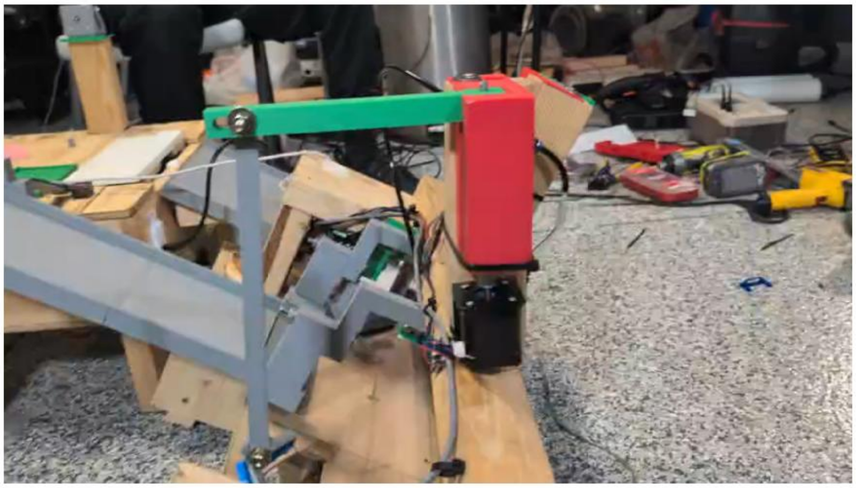
Demo (Partial Complete req. 5)



Demo (Partial Complete req. 5)



Demo (Partial Complete req. 5)

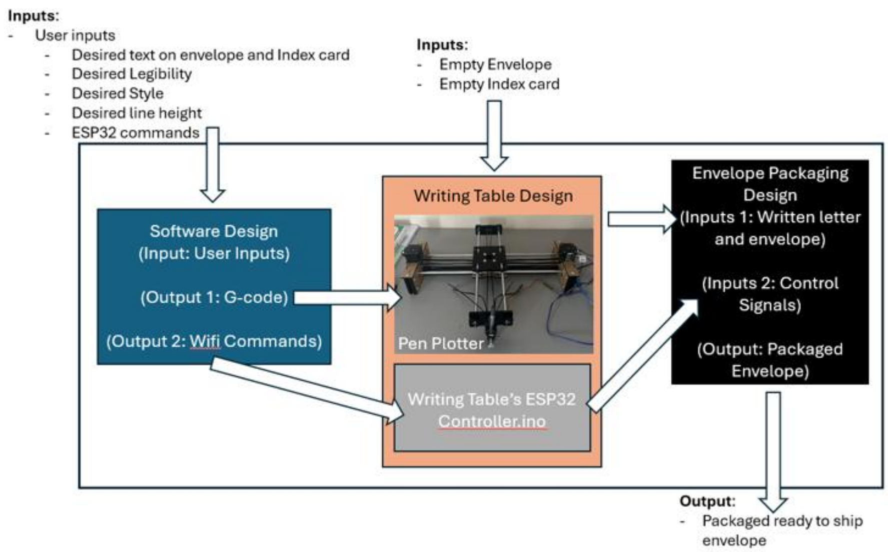


Demo (Completes req. 6)

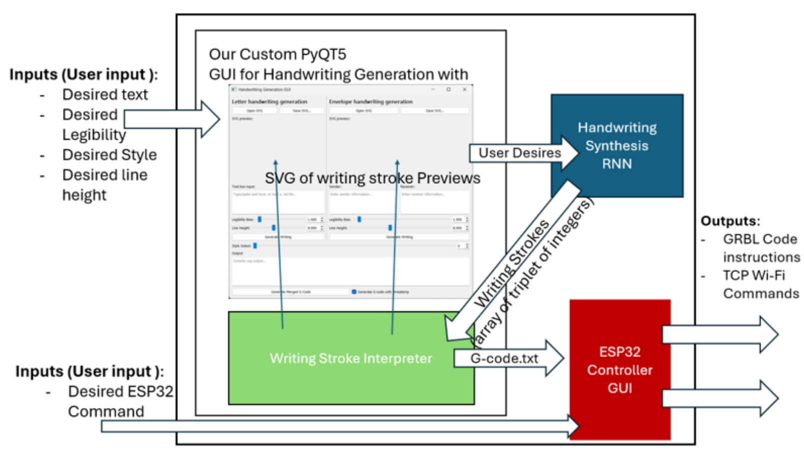
- IKRANBIRD AX4 Pen Plotter — **$129.99**
- 5 - Nema 17 Stepper Motors — **$55**
- 20 Pack - 624 Bearings — $**12.80**
- 4 Pack - Stepper Motor L-Shaped Mounting Bracket — **$8.99**
- 5 ESP32 boards — **$23.46**
- 5 Stepper Drivers / TMC2209 5-pack — **$22.88**
- 30 Pack Suction cups — **$6.99**
- 10 Meter air hose pipe — **$15.99**
- 5 Pack MG9S0 9G Servo — **$15.99**
- Teyleten Robot 24 Pins ATX Power Supply Adapter — **$9.99**
- 6 Pcs DC 3.3 Relay Module Channel Optocoupler Module — **$9.99**
- 2 - 12 V Mini DC Vacuum Pump — **$39.78**
- Hardware Fasteners (PLA, screws, nuts, washers, etc.) — roughly **$50**
- Wood — roughly **$80**

GRAND TOTAL: $481.85

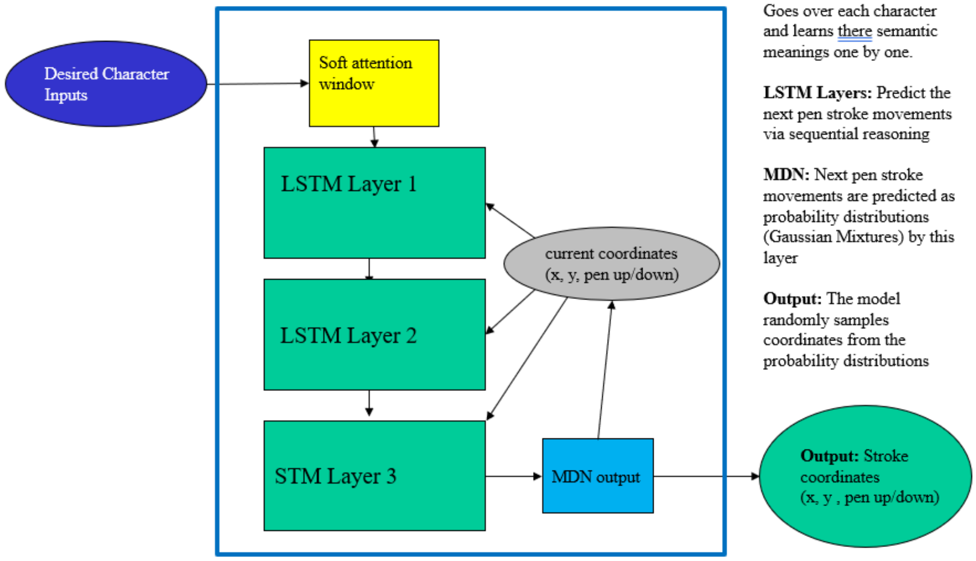
Design (High level overview)



Design (Software)



Design (Software: Neural network)



Soft Attention Window: Goes over each character and learns there semantic meanings one by one.

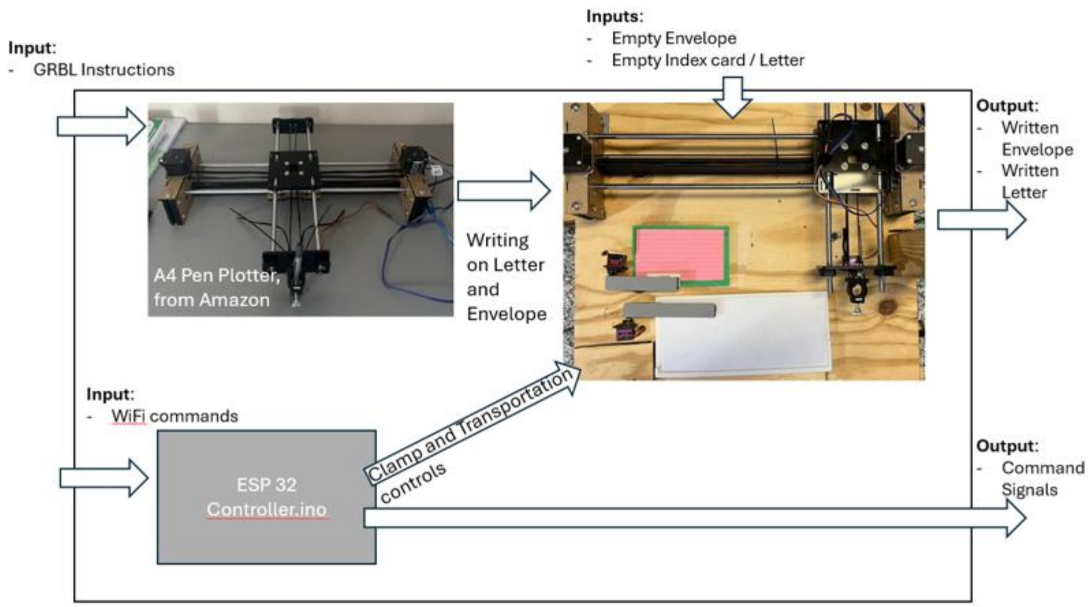
LSTM Layers: Predict the next pen stroke movements via sequential reasoning

MDN: Next pen stroke movements are predicted as probability distributions (Gaussian Mixtures) by this layer

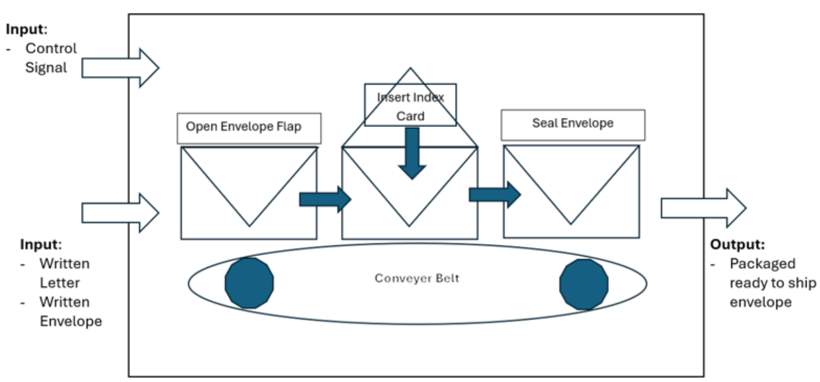
Output: The model randomly samples coordinates from the probability distributions

Design (Software: ESP32 Controller GUI)

Design (Writing Table)



Design (Envelope Packaging)



Applicable standards

- **G-code:** Standard language for CNC machine code
  - Our pen plotter uses G-code for plotting
- **USPS Mailing Standards:** Envelope and postage compliance
  - Our envelope uses a #10 envelope following this standard

Constraints

- **Cost:** Total system cost <= $500
  - Target Cost: <= $300
- **Time:** Must be designed, built, and tested within one academic semester
- **Human Interaction:** Maximum one human action per cycle
- **Physical Medium:** Must use real pen on standard #10 envelope and 3x5 index card
- **Power:** 120V, 15A outlets
- **Electronics**: electronics had questionable reliable during design

Trade offs

- Transfer Mechanism Trade-off
  - Original design used a suction cup system for precise envelope placement
  - Final design: Motor-driven lever lifts the platform, envelope slides into assembly line via gravity
  - Result: Simpler system within project constraints

- Stamping trade off
  - Original design used a suction cup system for precise envelope placement
  - Stamping step moved to immediately after the pen plotter

- Pre-Trained vs Trained ML model
  - Pre-trained model had flaws such as repeating characters, fading styles
  - Training the model on newer architecture solved these issues

Press Release

(Tampa, Florida, May 1, 2026) – Orion Mailing Solutions

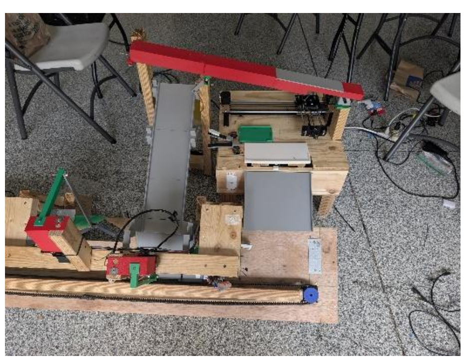
The team Orion Mailing Solution is pleased to announce that we designed and built an automated handwriting generation, letter writing and envelope packaging system for the company Orion Defense solution at the University of South Florida.

As AI replaces humans in many different sectors, human touch is lost in many aspects, handwriting being one. Human handwriting carries personality and authenticity in letters that printed and machine generated writing lacks. Which is why we addressed the issue of:

“Can we design and build an automatic continuous assembly line that can write on each envelope with a unique handwriting style, insert an index card, and stamp each envelope for mailing with minimal human involvement (one human in the loop) for less than $500, preferably less than $300 that cannot be detected as machine generated handwriting by at least 90% of people?”

To address this, we built an automated handwriting generation and packing machine. It consists of hardware, software and AI components that delivers an end-to-end automatic solution to authentic handwriting.

The AI component is a Recurrent Neural network that is capable of generating human handwriting. It features a software with an easy to use user interface, and a G-code conversion engine. The hardware part consists of a pen plotter that writes with a real pen, an assembly line for scaling and an automated envelope packing system.



Poster

Automated Handwriting generation and envelope packaging system

Aaryan Kumar, Abrar Zahin, Frankie Cacio, Jonathan Wong, Joshua Qoroth

UNIVERSITY of SOUTH FLORIDA

Background

AI is slowly phasing out human labor returning free time from repetitive tasks, however there is a loss in some human aspects of life, such as handwriting, that carry personal meaning and emotional connection.

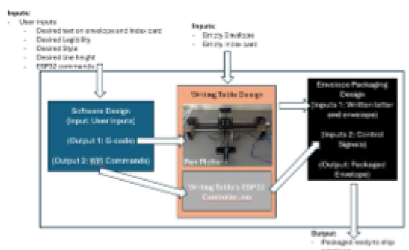
Problem

Can we design and build an automatic continuous assembly line that can write on each envelope with a unique handwriting style, insert an index card, and stamp each envelope for mailing with minimal human involvement (one human in the loop) for less than $500, preferably less than $300 that cannot be detected as machine generated handwriting by at least 90% of people?

Requirements

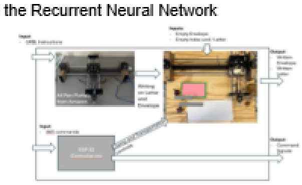
- As a user I want the handwriting and packaging system to be automated with at most 1 human interaction between the start (loading the index card and envelope) and the end (picking up the packaged envelope) so that I will be able to save time and be able to do other tasks as the envelope is being processed
- As a user I want the index card and the envelope to look human written with a real pen so that 90% of people will not be able to detect that it was written by a robot
- As a user, I want each envelope and index card to be written in a unique handwriting style so that no two handwritten outputs look identical across at least 1,000 jobs.
- As a user, I want the envelope to be packaged with an index card automatically so that I can save time and do other tasks

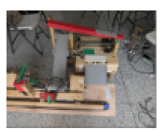
Design



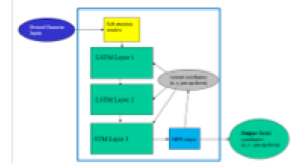
The design consists of Three parts: The Hardware, The Software and the Recurrent Neural Network

- Hardware:

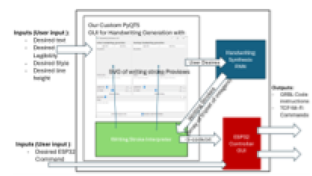




- RNN :



- Software:



Acknowledgements

- Cristian Dutescu (Orion Defense Solutions)
- Dr. Ken Christensen (University of South Florida)
- Alex Grave's Github repository for the AI model: https://github.com/sjvasquez/handwriting-synthesis
- AX4 Pen Plotter Writing Robot Open Source DIY Kit Corexy A4 Size Learn Programming: Bought from Amazon.com from IKRANBIRD store

Bellini College of AI, Cybersecurity and Computing CIS 4910 Senior Project – Spring 2026

Status Reports

Status Report for Orion Defense Solutions for Week # 3

What were the goals for this two-week period?

- Complete market research and identify the best pen plotter under $250. Have Final selection approved and order places by end of day Sunday January 25th.
- Select standard envelopes and 3x5 index cards and include them in the same order as the pen plotter.
- Research and identify at least one viable handwriting generation technique that can be used to generate a 100% unique, nonrepeating handwriting output.
- Create a rough functional block diagram of the software system that shows major components and data flow at a high level.
- Set up a communication system with our project sponsor for easy communication throughout the project.

What goals were accomplished in this two-week period?

- We successfully selected and ordered our pen plotter by Sunday January 25th.
- We successfully ordered standard envelopes and 3x5 index cards along with the pen plotter
- We successfully found 2 handwriting generation techniques that have paths to 100% unique handwriting output
- We successfully sketched a rough functional block diagram of the potential software system
- We successfully set up a group message using the Signal communication app for easy communication with our project sponsor.

Reflect critically on any goals not accomplished.

These last 2 weeks we were able to meet our goals that we set because we were in the research phase. We successfully completed our research into our pen plotter and handwriting techniques, so we are on pace with our project plan. While we have not made any physical progress on implementation for that no plans have been made yet, we have set ourselves up to continue making progress on achieving our end goal by the end of semester.

What are the goals for next two weeks?

- Complete assembly of pen plotter
- Select one of the two viable handwriting techniques and work on implementing them to create 100% unique handwriting - Dip our toes into the previously shown works. Begin on implementing an initial protype - Select 3D modeling software (Onshape or SolidWorks) and begin learning the basics.
- Develop an initial mechanical concept for transferring the card from the pen plotter to the envelope system.
- Research and select components for envelope opening, including a basic vacuum system.
- Figure out stamp integration.

How many hours were spent on each goal noted above in the past two weeks?

<u>Frankie Cacio:</u>

- [Completed] Researched possible pen plotters (**Total**: ~2hrs)
- [Completed] Researched potential handwriting generation techniques (**Total**: ~45min)

<u>Jonathan Wong</u>:

- [Completed] Researched possible pen plotters and handwriting generation techniques (**Total**: ~1 hr)
- [Completed] Finding existing documents for selected pen plotter (AX4) and Software to visualize G-Code (Candle, https://github.com/Denvi/Candle) (**Total**: ~45 min)

<u>Abrar Zahin</u>:

- [Completed] Found VRNN handwriting synthesis technique (https://github.com/emreaksan/deepwriting) (**Total**: ~1 hr)
- [Completed] Found LSTM-RNN for handwritting generation (https://github.com/sjvasquez/handwriting-synthesis) (**Total**: ~1 hr)

- [Completed] Designed high level architecture for both approaches for handwriting synthesis to robot generating physical letter (**Total**: ~1 hr)

 <u>Aaryan Kumar and Joshua Ooroth:</u>

- [Completed] Research existing envelope inserting machines and reference videos (Total: ~0.5 hrs)
- [Completed] Evaluating feasibility and cost of commercial machines vs. DIY approach (Total: ~0.5 hrs)
- [In Progress] Brainstorming conveyor belt and 3D-printed clamp concepts and researching envelope opening methods using vacuum system (Total: ~0.5 hrs)
- [In Progress] Comparing 3D modeling software options (Onshape vs. SolidWorks) (Total: ~0.5 hrs)

Status Report for Orion Mailing Solutions for Week #5

What were the goals for this two-week period?

1. Complete assembly of pen plotter
2. Select one of the two viable handwriting techniques and work on implementing them to create 100% unique handwriting 3. Dip our toes into the previously shown works. Begin implementing an initial prototype
4. Select 3D modeling software (Onshape or SolidWorks) and begin learning the basics.
5. Develop an initial mechanical concept for transferring the card from the pen plotter to the envelope system.
6. Research and select components for envelope opening, including a basic vacuum system.
7. Figure out stamp integration

What goals were accomplished in this two-week period?

1. We completed assembling the pen plotter and successfully ran a test.
2. We were able to select one handwriting technique (RNN-handwriting synthesis)
3. We were able to implement 3 of the 5 components (plotter, frontend app, handwriting model) for the initial prototype
4. We were able to begin designing our 3D printed parts with Onshape (we designed and printed one of the parts for the vacuum system)
5. We were able to find 8 parts to put together a prototype vacuum system.

Reflect critically on any goals not accomplished.

1. We are still brainstorming and debating over ideas for transferring the card from the pen plotter to the stuffing system.
2. Since the assumptions from our client allow one human in the loop, we are debating the idea of having the human work on the stamping part.

What are the goals for next two weeks?

1. Test a prototype with the 3 prototype components (frontend + AI + Pen plotter)
2. Test training the AI model for better, more human handwriting generation and test the alternative AI model
3. Finish a full sketch of the 3D modelling parts for printing.
4. Acquire the 8 parts for the vacuum system and test them.

How many hours were spent on each goal noted above in the past two weeks?

Frankie Cacio:

- [Completed] Assembled the pen plotter (Total: 6 hours)

- [Completed] Tested the pen plotter (Total: 2 hours)

Jonathan Wong:

- [Completed] Designed and implemented the frontend application + the G-code conversion script (Total: 7) - [Completed] Tested the pen plotter (Total: 2 hours)

Abrar Zahin:

- [Completed] Researched and designed more alternative handwriting synthesis techniques (Total 4 hours) - Attempted fine tuning the current handwriting synthesis technique on google colab (Total 3 hours)

Joshua Ooroth:

-[Completed] Researched and found 8 parts after designing the vacuum system (Total 3 hours)

-[Completed] Set up and test/debug 3D printer (2 hours)

- Initial design of the vacuum system

Aaryan Kumar:

- [Completed] Assembled the pen plotter (Total 3 hours)
- [Completed] Completed video tutorials for 3D printing + sketched a 3D model of a tube (vacuum system part) using OnShape (Total 6 hours)

Status Report for Orion Defense Solution for Week #8

What were the goals for this two-week period?

< Enumerate your goals for this past two-week period. Be sure that your goals are measurable and connected to your requirements. >

1. Improve the User interface, make it simpler by reducing the number of buttons. Automatically call the of g-code generation after writing stroke generation, combining two buttons into one.
2. Improve User interface, make user input more explicit. User inputs such as sender and receiver of envelope input text box, split the user input from one textbox into two. Combine the two individual writing style select input to one so that there are less repetitive user inputs. Combine timestamp sourcing from two buttons to one, before letter and envelope generation had individual timestamp checkboxes, now only one checkbox.
3. Design new function of g-code generation, combining letter and envelope g-code into one. Create a final output g-code that includes both letter and envelope g-code
4. When generating writing use multithreading instead of single thread so GUI doesn’t go unresponsive when writing is being generated. Have three threads for each writing generation, letter, sender, and receiver.
5. Test training the AI model for better, more human handwriting generation and test the alternative AI model
6. Finish a full sketch of the 3d modelling parts for printing

What goals were accomplished in this two-week period?

< Enumerate what goals were completed in this past two-week period. >

1. Simplification of User Input removed repetitive user inputs. Combined letter and envelope writing generation and g-code generation buttons into one button. Combined individual style selection for letter and envelope generation into one slider which picks for both.
2. Separated envelope text input for sender and receiver section
3. Merging letter and envelope g-code function
4. SVG to writing strokes function
5. New user input to control line height of writing generation
6. Using multithreading for writing generation
7. Training the AI model for better handwriting was accomplished.
8. We have 3d modelled 75% of the parts for printing

Reflect critically on any goals not accomplished.

< Explain why you did not complete any missed goals and what you plan to do moving forward with respect to these goals. >

1. *SVG to writing stroke generation is broken for envelope Load saved SVG, caused by separation of sender and receiver input for envelope. Original design was not designed to handle multiple svg groups. Moving forward a new function will be written for the envelope Load SVG which extracts the group and will update sender and receiver strokes respectively, generation of the new combined sender receiver SVG has already separated into group and ID'ed sender and receiver when creating the combined SVG.*

2. *Testing the alternative AI model (the Vision Transformer). Currently, we've decided on training and improving the current AI model we're using. We're currently optimizing the model by doing a hyperparameter grid search for better hyperparameter combinations, so that the model generates handwriting that looks more human (consistent letter structures and spacing).*
3. *We were only able to model 75% of 3d printed parts due to some constraints such as redesigning previous models for the design changes we are undergoing. We changed how the carrier system works, and this required redesigning the tube, envelope carrier, and the opening knife.*

What are the goals for next two weeks?

< Enumerate your goals for next two weeks. Be sure that your goals are measurable and connected to your requirements. >

1. Documentation of GUI code and how to use it. Documentation should include what each function and file does or purpose, how to use the GUI form a high level, specification of which buttons connect to which function, and troubleshooting steps.

2. Fix the load svg function for envelope. Function needs to separate and interpret the saved svg to writing strokes, try to find the key id's of the svg and if not found should throw an error.
3. Modify the generation of final g-code (the g-code which contains both letter and envelope writing). Need flip the envelope writing upside down.
4. Finish hyperparameter Grid search on the RNN model and find how much the sampled handwriting styles affect the model output, and how the style NumPy files are generated in the first place.
5. Modeling 100% of 3d printed parts and testing the vacuum system.

How many hours were spent on each goal noted above in the past two weeks?

< For each person on the team list the goals they worked on including completed and non-completed goals

Jonathan Wong:

1. Simplification of User Input removed repetitive user inputs. Combined letter and envelope writing generation and g-code generation buttons into one button. Combined individual style selection for letter and envelope generation into one slider which picks for both. (2 hours)
2. Separated envelope text input for sender and receiver section (2.5 hours)
3. Merging letter and envelope g-code function (1.5 hours)
4. SVG to writing strokes function (1 hour)
5. New user input to control line height of writing generation (0.5 hours)
6. Using multithreading for writing generation (1 hour)

Abrar Zahin:

Active work:

1. Fully Trained the model in 3 training sessions. Change the hyperparameter for each training session, record the results
(failures such as loss overshoots to NaN and infinity for unstable hyperparameter values). (~ 1 hours)
2. Debug crashing training sessions (first issue: batch size and learning rate mismatch, second issue: memory overload due to large batch sizes) (3 hours)
3. Perform an inference study (from 3 training sessions), create a report (4 hours)

Passive monitoring:

4. Monitor model training (whether loss is stable or not), monitor the GPU cluster for memory overload and temperature
(~15 hours)

Frankie Cacio

1. Designed templates for writing platform (1 hour)
2. Built box/platform for pen plotter and index card/envelope positions, with space inside for electronics and wiring (4 hours)
3. Designed lifting mechanism parts to transfer envelope from pen plotter to assembly line (3 hours)

Aaryan Kumar:

1. Modelled Stamping Mechanism and printed it (5 hours)
2. Modelled Envelope Carrier and printed it (3 hours)
3. Modelled Envelope Opener but needs redesigning (2 hours)
4. Finalized the planning of the envelope packing machine (1 hour)

Joshua Ooroth:

1. Completed engineering specifications for envelope carrier system, and made modifications to account for unforseen limitations (3 hours)
2. Made paper prototypes for modeling parts (2 hours)
3. Obtained timing belt and designed an appropriate gear to use with a stepper motor (1 hour)

Status Report for <u>Orion Defense Solutions</u> for Week #<u>13</u>

What were the goals for this two-week period?

1. Implement interface to communicate actions with controller via WI-FI

2. Complete automation for transfer system of envelope from pen plotter system to conveyer belt

3. Finish hyperparameter Grid search on the RNN model and find how much the sampled handwriting styles affect the model output, and how the style NumPy files are generated in the first place.

4. Design and implement clamp to hold index card and envelope in place during writing
5. Design system to transfer index card to envelope insertion point
6. Implement vacuum system to open envelope for index card to be inserted
7. Remodel 3d models for better optimization from the demo

What goals were accomplished in this two-week period?

1. Finish hyperparameter Grid search on the RNN model and find how much the sampled handwriting styles affect the model output, and how the style NumPy files are generated in the first place.
2. Implement interface to communicate actions with controller via WI-FI
3. Complete automation for transfer system of envelope from pen plotter system to conveyer belt
4. Design and implement clamp to hold index card and envelope in place during writing
5. Implement vacuum system to open envelope for index card to be inserted **Reflect critically on any goals not accomplished.**

1. Index card transfer system has yet to be implemented due to needing more parts and needing to make necessary adjustments to other systems to be completed. However, we have created a design that just needs to be implemented.

What are the goals for next two weeks?

1. Implement newer neural network techniques on the model for potential improvements. Two techniques being implemented: a. Switch to Style Embedding from hidden state embedding
   b. Attempt implementing Variational Auto Encoder.
2. Finalize index card transfer mechanism
3. Finalize Stamp Mechanism
4. Finalize wiring for hardware
5. Optimize for repeatability and reliability

How many hours were spent on each goal noted above in the past two weeks?
Abrar Zahin:

Finish hyperparameter Grid search on the RNN model and find how much the sampled handwriting styles affect the model output, and how the style NumPy files are generated in the first place: (

- Hyperparam grid serach (~12-14 hours)
- Extract Style (4 – 5 hours)
- New technique research (3 hours)
- Legibility and inference study (6 hours) **Frankie Cacio:**
- Completed automation of envelope transfer system (5 hours)
- Designed clamp and implemented clamp automation (5 hours)
- Fixed alignment of pen plotter origin and index card positioning for more consistent and accurate results when plotting (1 hour)
- Index card mechanism planning (2 hours) **Joshua Ooroth:**
- Finalized design for index card system (3 hours)
- Placement and wiring for stepper motors for opening and sealing systems (3 hours)
- Modifications to physical mechanisms for envelope carriage to optimize reliability. (2 hours) **Aaryan Kumar:**
- Finished optimizing 3d models from the demo (6 hours)
- Index card mechanism planning (2 hours)
- Modeling Index card mechanism parts (2 hours) **Jonathon Wong:**
- Implemented a listener for candle to begin trigger for packaging (5 hours)
- Created a controller interface to manually trigger specific actions (5 hours)
- Worked on wiring for power supply and ESP32 controller (4 hours)

New Knowledge/Skills Gained

Frankie Cacio:

I learned a wide range of hands-on engineering skills throughout this project, including electrical wiring, motor control, servo control, ESP32 programming, relay control, and basic troubleshooting for embedded hardware. I also learned more mechanical design skills, such as 3D modeling custom brackets and mounts, 3D printing functional parts, and adjusting designs based on real-world fit and performance. A lot of this knowledge came from trial and error, testing individual subsystems, researching components, reading documentation, and working through problems as they came up during the build.

Jonathan Wong:

This project provided valuable experience in connecting software with physical hardware systems. I learned practical GUI development skills, especially for interfaces that communicate with external devices such as Arduino, ESP32, and GRBL-based controllers. I also gained a better understanding of how to use LLMs effectively by first creating a strong program structure, then breaking the project into smaller, manageable components that could be developed and tested individually. In addition, the project improved my understanding of electrical and hardware challenges, including signal noise, grounding issues, and interference, which can affect communication between devices. Working with the ESP32 also helped me better understand the strengths and limitations of microcontrollers: they are useful for lightweight control tasks, but larger systems with many simultaneous demands may require more robust hardware or a more distributed control approach. Overall, this project gave me a more realistic understanding of how software, electronics, and mechanical systems must work together in an automated system.

Joshua Ooroth:

Through this project, I learned the importance of compartmentalization and breaking a large project into smaller, manageable subgoals. One major lesson was that combining too many components at once increased the overall risk level and made troubleshooting much more difficult than if each section had been developed and tested separately. I also realized the value of having a wider variety of electronic components available and gaining more knowledge about how different parts function. Another takeaway was understanding the tradeoff between cost and quality, since choosing cheaper components can reduce reliability and performance. In terms of mechanical design, I felt confident in my abilities, though refining designs took time and required clear communication and collaboration with teammates to successfully share and develop ideas.

Aaryan Kumar:

Through this project, I gained valuable new knowledge and skills across multiple areas of engineering. I learned a significant amount about mechanical engineering through 3D modeling in Onshape, which helped me better understand how to design and visualize parts for real-world applications. I also learned the importance of combining different branches of a project and how mechanical, electrical, and design components must work together as a complete system. One key lesson was realizing the value of testing each individual part before integrating everything, since identifying problems early makes troubleshooting much easier. Additionally, I gained experience with electrical engineering concepts, especially wiring and the troubleshooting process that comes with ensuring circuits function correctly. Overall, this project strengthened both my technical skills and my understanding of how different engineering disciplines connect in a practical setting.

Abrar Zahin:

I learned a lot about Recurrent Neural Network architectures. Through contrinuous Research, Implementation, and testing, I learned how different RNN techniques affect model outputs. I have a deeper understanding of how RNN’s work with LSTMs as a state machine learning architecture.

Through this project I learned how to choose an AI model for a task through research and comparison and then how to develop it iteratively through testing different research techniques. I tried several novel techniques and while some of them worked to make the model better, some of them did not. I learned how to predict which techniques will work and which wont. Additionally, I learned how ML models work with frontend applications like our current GUI.